

Álvaro Torres Carrasco

Software Engineer (Backend)

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SUMMARY

Backend Software Engineer with 5+ years in production with Java (Quarkus), Node.js and Python. I build systems end to end: microservices, APIs, job queues, notifications and the infrastructure they run on. Looking for a remote backend role.

TECHNICAL SKILLS

Programming languages: Java, TypeScript, Python, JavaScript, SQL

Frameworks: Quarkus, Spring Boot, Node.js, NestJS, Django, Django REST Framework, Drools

Data and messaging: PostgreSQL, Oracle, MongoDB, Redis, Apache Kafka, Celery, Prisma

APIs and integration: REST, GraphQL (Apollo, Graphene), WebSockets, OAuth 2.0, JWT

Cloud and DevOps: Docker, Kubernetes, Amazon Web Services (AWS), GitHub Actions, GitLab CI, Jenkins, Caddy, Proxmox

Testing, tools and methodologies: JUnit, Pytest, Jest, Vitest, Git, Linux; Agile, Scrum, Kanban

Languages: Spanish (native), English B2 (Trinity College London)

PROFESSIONAL EXPERIENCE

Software Engineer (Backend), Soltel – Seville (remote) 01/2025 – present

- Develop more than 10 Java (Quarkus) microservices that unify several public grant and scholarship systems.
- Built from scratch, with Drools, the case-scoring system: intake, scoring and resolution in production.
- Migrating a shared data bus to microservices with an event-driven architecture (Kafka) and proxies (strangler fig pattern); data modelling in Oracle and MongoDB, tests with JUnit and CI/CD with GitLab CI and Jenkins.

Career break, Own entrepreneurial project 01/2024 – 12/2024

Software Engineer (Backend), Therapyside – Madrid (remote) 03/2022 – 01/2024

- Maintained the Django REST Framework backend of 5,000+ daily active users and migrated it from Python 2.7 to 3.7.
- Raised test coverage from 12% to over 85% and set up the Pytest structure from scratch.
- Redesigned real-time and push notifications (Celery, Redis, Django Channels) for thousands of concurrent users.
- Reduced PostgreSQL load with indexes, by removing N+1 queries, Redis caching and documents in MongoDB.

Software Engineer (Backend), Z1 Digital Studio – Seville (remote) 03/2020 – 03/2022

- Developed backends for clients in the US and Canada: Storytelling with Data, Waking Up, energy and social networking.
- Built REST and GraphQL APIs (Django, Graphene, NestJS, Apollo), a NestJS and MongoDB back office and OAuth login.
- Deployed on AWS with Docker and CI/CD on GitHub Actions, with test coverage above 80%.

PROJECTS

Job queue and notification service In development

- TypeScript queue on PostgreSQL, in production in Quedamos and Huellas: transactional enqueueing, SKIP LOCKED, retries with backoff, duplicate-free cron jobs and workers with configurable concurrency.
- Notification microservice (push and email) with idempotency, per-channel adapters and a dead-letter queue.

Huellas · github.com/alvarotorresc/huellas In development

- Platform to reunite lost pets with their families: weighted, explainable rule-based matching between sightings and listings, with proximity search in PostGIS (geography type, GiST index, ST_DWithin).
- Critical alerts with their own acknowledgement (single-use token) and an email fallback if no acknowledgement arrives within 3 minutes, duplicate-free thanks to the outbox pattern and to the expiry date of each push.
- Rate limits per action and per origin, uniqueness constraints enforced in the database itself rather than in the application, and encrypted contact details, with an audit log entry for every disclosure.

Quedamos · quedamos.alvarotc.com In production

- App for organising group meetups, with real-time sync between members; NestJS API with Prisma and PostgreSQL, JWT verified with JWKS, push with Firebase, helmet, rate limiting and Sentry.
- 608 tests with a minimum coverage of 80%; CI/CD that validates the migrations, publishes the image to Docker (ghcr.io) and runs a smoke test after every deploy, checking that the freshly published image is running.

Self-hosted infrastructure (VPS and homelab) In production

- About ten services on Docker Compose behind Caddy, including the Quedamos API: Hetzner VPS and Proxmox VE homelab (LXC, VMs and GPU passthrough), with only WireGuard exposed, a DNS-01 wildcard certificate and fail2ban.
- Own monitoring (systemd timers, Telegram alerts and an external heartbeat), daily backups with restic (7 daily, 4 weekly, 6 monthly) and pg_dump, with tested restores, and nightly VM backups with vzdump.

EDUCATION

IES Velázquez, Higher Technician in Web Application Development – Seville 2018 – 2020